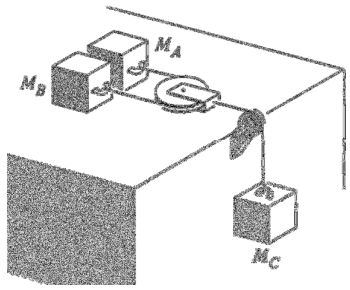
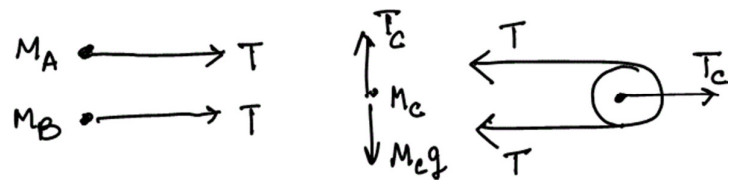
PH101: Tutorial-2

Newton's laws of motion-I

Problem 1: Two masses, M_A and M_B, lie on a frictionless table as shown in Fig. They are attached to either end of a light rope of length l_1 which passes around a pulley of negligible mass. The pulley is attached to a rope of length l_2 connected to a hanging mass, M_C. Find the tensions in the ropes.





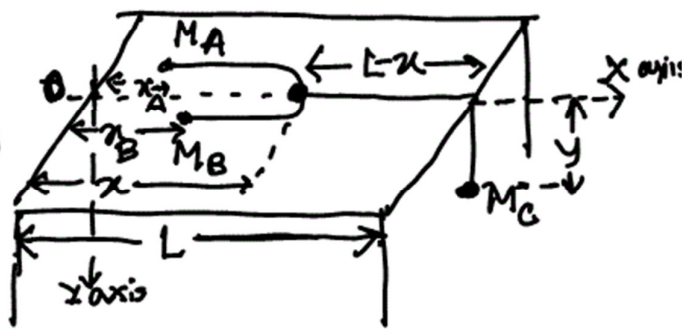
Constraints:

$\cdot\ (x - x_A) + (x - x_B) = l_1$

$\cdot\ L - x + y = l_2$

$\therefore\ 2\ddot{x} = \ddot{x}_A + \ddot{x}_B$

& $\ddot{x} = \ddot{y}$



Eqs of motion:

$T = M_A \ddot{x}_A \quad | \quad T = M_B \ddot{x}_B \quad | \quad M_C g - T_C = M_C \ddot{y}$

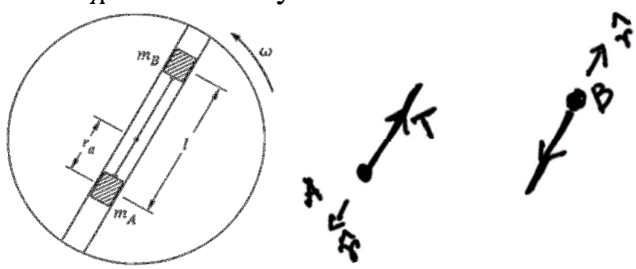
Also $T_C = 2T$.

Solving we get,

$$\boxed{T = \frac{2 M_A M_B M_C g}{M_C (M_A + M_B) + 4 M_A M_B}}$$

and $T_C = 2T$.

Problem 2: A disk rotates with constant angular velocity ω, as shown in Fig. Two masses, m_A and m_B, slide without friction in a groove passing through the center of the disk. They are connected by a light string of length l, and are initially held in position by a catch, with mass m_A at distance r_A from the center. Neglect gravity. At $t = 0$ the catch is removed and the masses are free to slide. Find $\ddot{r}_A$ immediately after the catch is removed in terms of m_A, m_B, l, r_A, and ω.



For m_A

$$-T = m_A(\ddot{r}_A - r_A\dot{\theta}^2) = m_A(\ddot{r}_A - r_A\omega^2)$$

m_B

$$-T = m_B(\ddot{r}_B - r_B\omega^2).$$

where $r_B = l - r_A$

$$\Rightarrow \ddot{r}_B = -\ddot{r}_A$$

Solving,

$$\boxed{\ddot{r}_A = r_A\omega^2 - \frac{l\omega^2}{1 + m_A/m_B}}$$

Problem 3: A mass m is connected to a vertical revolving axle by two strings of length l, each making an angle of 45° with the axle, as shown. Both the axle and mass are revolving with angular velocity ω as shown in Fig. Gravity is directed downward. Draw a clear force diagram for m. Find the tension in the upper string, T_{up}, and lower string, T_{low}.

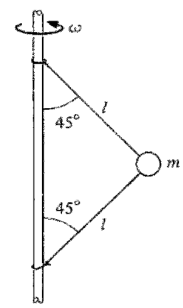
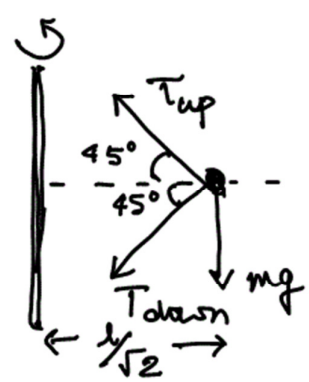


Fig.3

a)

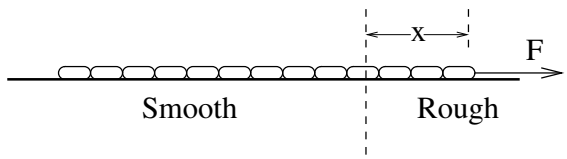


b) using Newton's 2nd law;

- $\frac{T_{up}}{\sqrt{2}} - \left(mg + \frac{T_{down}}{\sqrt{2}}\right) = 0$
- $\frac{T_{up} + T_{down}}{\sqrt{2}} = m\frac{l}{\sqrt{2}}\omega^2.$

$$\boxed{\begin{aligned} T_{up} &= \frac{ml\omega^2}{2} + \frac{mg}{\sqrt{2}} \\ T_{down} &= \frac{ml\omega^2}{2} - \frac{mg}{\sqrt{2}} \end{aligned}}$$

Problem 4: A heavy chain of length L and mass ρ per unit length is pulled along a horizontal surface consisting of a smooth section and a rough section by a constant force $\vec{F}$ as shown in Fig. If the chain is initially at rest on the smooth surface with x=0, and if the coefficient of friction between the chain and the rough surface is μ, determine the velocity of the chain when x=L.



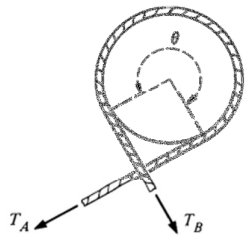
|friction force on length x| $= \mu \rho x g$.
(acting toward left)

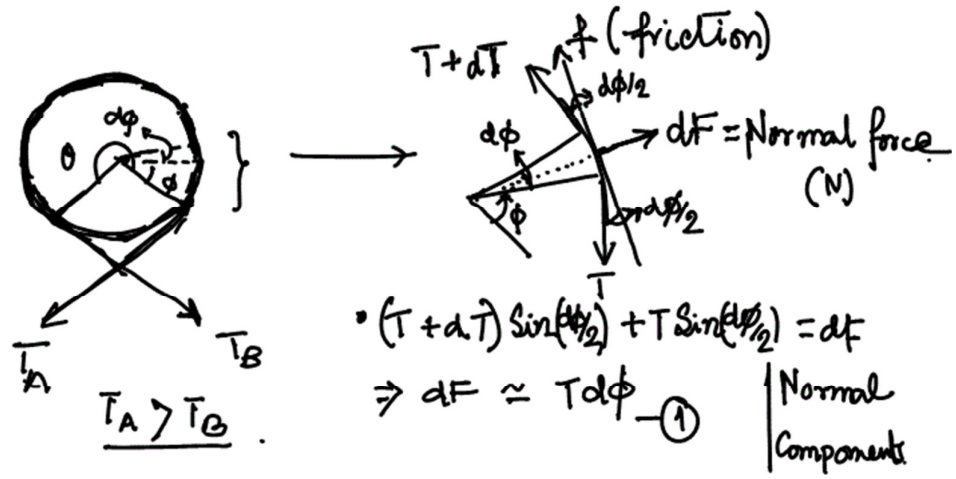
Eq of motion: $F - \mu \rho x g = (\rho L) v \frac{dv}{dx}$

or $\frac{d}{dx}\left(\frac{v^2}{2}\right) = (F - \mu \rho x g)/\rho L$

$\therefore \boxed{v = \sqrt{\frac{2F}{\rho} - \mu g L}}$.

Problem 5: A device called a capstan is used aboard ships in order to control a rope which is under great tension. The rope is wrapped around a fixed drum as shown in Fig. The load on the rope pulls it with a force T_A, and the sailor holds it with a much smaller force T_B. Show that $T_B = T_A e^{-\mu\theta}$, where μ is the coefficient of friction and θ is the total angle subtended by the rope on the drum.





$$\bullet\ (T+dT)\sin\left(\frac{d\phi}{2}\right) + T\sin\left(\frac{d\phi}{2}\right) = dF$$

$$\Rightarrow dF \simeq T\,d\phi \quad \text{— (1)}$$

• tangential forces are in equilibrium,

$$(T+dT)\cos\frac{d\phi}{2} - T\cos\frac{d\phi}{2} + f = 0.$$

$$\Rightarrow f \simeq -dT \longrightarrow (2)$$

Now $f \leq \mu N$.

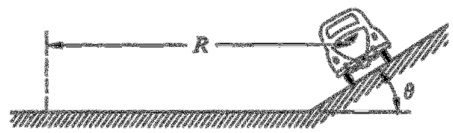
$\therefore f = \mu\, dF$, so from (1) & (2);

$$dT = -\mu T\, d\phi$$

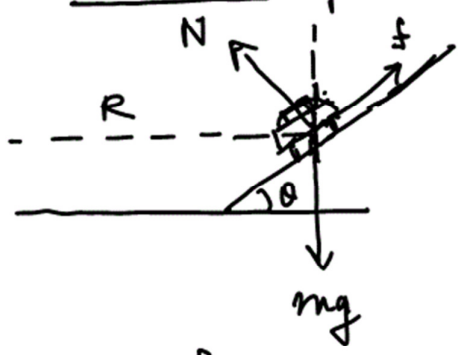
$$\int_{T_A}^{T_B} \frac{dT}{T} = -\mu \int_0^{\theta} d\phi = -\mu\theta$$

$$\therefore \boxed{T_B = T_A e^{-\mu\theta}}$$

Problem 6: An automobile enters a turn whose radius is R as shown in Fig. The road is banked at angle θ, and the coefficient of friction between wheels and road is μ. Find the maximum and minimum speeds for the car to stay on the road without skidding sideways.



I. Minimum speed Case:



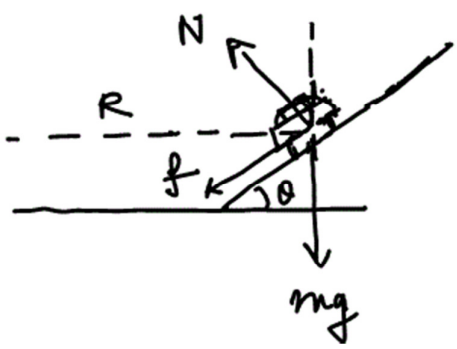
$$N\cos\theta + f\sin\theta - mg = 0$$

$$\& \quad N\sin\theta - f\cos\theta = \frac{mv^2}{R}.$$

$$f \le \mu N.$$

With $f_{max} = \mu N \Rightarrow v_{min} = \left[Rg\left(\frac{\sin\theta - \mu\cos\theta}{\cos\theta + \mu\sin\theta}\right)\right]^{1/2}$

II. Maximum speed case:



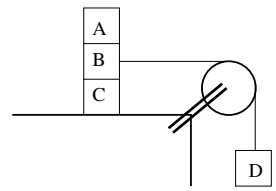
$$N\cos\theta - f\sin\theta - mg = 0$$

$$\& \quad N\sin\theta + f\cos\theta = \frac{mv^2}{R}.$$

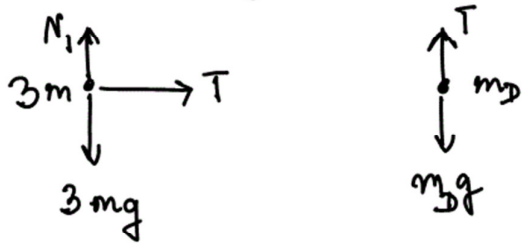
$$f \le \mu N.$$

$$v_{max} = \left[Rg\left(\frac{\sin\theta + \mu\cos\theta}{\cos\theta - \mu\sin\theta}\right)\right]^{1/2}$$

Problem 7: Three blocks A, B, and C of equal masses m each are placed on top of each other on a frictionless table as shown in Fig.7. Another block D of mass m_D is attached with the block B through an inextensible string over a pulley. The coefficients of static and kinetic frictions between A, B and B, C are $\mu = 0.5$.
(a)Find the maximum mass m_D* of block D so that all three blocks move together without slipping over each other. (b) If the mass of the block D is just greater than m_D* which block would slip first with respect to B? (c) For what value of m_D will both A and C start slipping?



a) Three blocks together:



$N_1 = 3mg$ $\qquad$ $m_D g - T = m_D a$.

$T = 3ma$

$$\Rightarrow \boxed{a = \frac{m_D g}{3m + m_D}}$$

Friction force

On A : $f_A^{max} = \mu m g$

On C : $f_C^{max} = 2\mu m g$

$\rightarrow$ A will slip first.

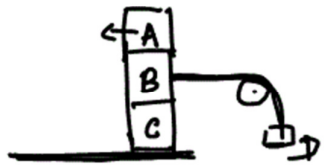
Condition of no slipping:

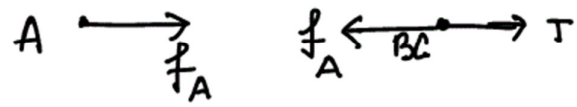
$$m \cdot \frac{m_D g}{3m + m_D} \leq \mu m g$$

or $m_D^{max} = \dfrac{3m\mu}{1-\mu} = 3m.$

b) A will slip first.

c) assume that A starts to slip; but BC moves together





$\cdot\ f_A = \mu m g$

$\cdot\ T - f_A = 2m\, a_{BC}$ | $m_D g - T = m_D a_D$

$= m_D a_{BC}$

or $a_{BC} = \left(\frac{m_D - \mu m}{2m + m_D}\right) g.$

Condition of <u>slipping of C</u>:

$$m a_{BC} > f_C = \mu \cdot 2mg$$

or $\frac{m_D - \mu m}{2m + m_D} > 2\mu$

or $\boxed{m_D - \frac{m}{2} > m_D + 2m}$ (with $\mu = \frac{1}{2}$)

This condition can never be satisfied, so C will not slip.